

Merrivoss Bioanalysis Inc

TP04 DDI Assessment with CYP3A4 Confirmation

Final study report R1

Study MV-2016-095

Accession M16/BIO/02193

Prepared for Tervemere Pharma AG

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1 Study scope and identification

This final report concerns the TP04 test article set for Tervemere Pharma AG. The study comprised 18 compounds and opened on 05/30/2016. The interim communication was dated 06/07/2016; this R1 report closes interpretation on 06/16/2016.

Table 1. Study scope and DDI benchmarks

Metric	Value
Project	MV-2016-095
Client	Tervemere Pharma AG
Compounds tested	18
Series / chemotype	Tervemere aminopyrazolopyrimidine series 04 (Aminopyrazolopyrimidine)
Assays	CYP inhibition panel (1A2/2C9/2C19/2D6/3A4) at 10 µM, CYP3A4 inhibition IC50
Turnaround (calendar days)	17
Price	USD 20,300
CYP1A2 >50% inhibition @10 µM	2
CYP2C9 >50% inhibition @10 µM	4
CYP2C19 >50% inhibition @10 µM	5
CYP2D6 >50% inhibition @10 µM	2
CYP3A4 >50% inhibition @10 µM	5
Compounds with any CYP >50%	5
Ketoconazole (3A4) IC50 (µM)	0.035

The Five-isoform DDI screen and CYP3A4 inhibition curves answer related but distinct questions. Single-concentration inhibition identifies follow-up concerns; concentration-response work defines the CYP3A4 interaction under its probe-substrate conditions. Neither is a measured clinical interaction.

2 Five-isoform DDI screen

The human liver microsomal panel assessed CYP1A2, CYP2C9, CYP2C19, CYP2D6 and CYP3A4 at 10 μ M test article. Values below are percentage inhibition of probe-substrate activity under the screen conditions.

Table 2. Five-isoform inhibition at 10 μ M

Compound	CYP1A2 (%)	CYP2C9 (%)	CYP2C19 (%)	CYP2D6 (%)	CYP3A4 (%)
TP04-02700	-9	-4	5	-12	-2
TP04-02722	24	90	69	38	90
TP04-02724	45	12	8	31	39
TP04-02734	45	7	8	3	4
TP04-02736	1	10	3	8	3
TP04-02742	25	8	2	8	45
TP04-02746	83	49	87	78	71
TP04-02758	34	60	80	76	71
TP04-02784	7	7	-7	-2	-2
TP04-02808	10	6	-4	-7	2
TP04-02818	3	-2	10	-0	3
TP04-02820	4	-2	-2	2	14
TP04-02824	-0	8	-1	1	6
TP04-02850	-6	8	7	1	0
TP04-02860	8	100	61	19	57
TP04-02872	0	3	-0	-5	-3
TP04-02874	5	-2	-3	-2	10
TP04-02880	61	79	87	34	84

Single concentration 10 μ M, duplicate incubations, mean reported; negative values = no inhibition

Controls (IC₅₀, μ M): furafylline (CYP1A2) 2.1; sulfaphenazole (CYP2C9) 0.48; ticlopidine (CYP2C19) 1.1; quinidine (CYP2D6) 0.065; ketoconazole (CYP3A4) 0.035

The screen uses a common test concentration for comparison, not an IC₅₀ estimate for each isoform. Negative inhibition values remain in the released table; they are not reclassified as a demonstrated activation mechanism.

3 CYP3A4 confirmation table

CYP3A4 inhibition curves used midazolam 1'-hydroxylation as the probe reaction. The reported IC₅₀ is expressed in μ M. Bound entries remain bound statements rather than exact potency estimates.

Table 3. CYP3A4 concentration-response results

Compound	IC ₅₀ (μ M)	Probe substrate
TP04-02700	>50	midazolam
TP04-02722	1.3	midazolam
TP04-02724	16	midazolam
TP04-02734	>50	midazolam
TP04-02736	>50	midazolam
TP04-02742	14	midazolam
TP04-02746	4.7	midazolam
TP04-02758	3.7	midazolam
TP04-02784	>50	midazolam
TP04-02808	>50	midazolam
TP04-02818	>50	midazolam
TP04-02820	>50	midazolam
TP04-02824	>50	midazolam
TP04-02850	>50	midazolam
TP04-02860	6.8	midazolam
TP04-02872	>50	midazolam
TP04-02874	>50	midazolam
TP04-02880	0.71	midazolam

7-point curve, top 50 μ M; ketoconazole control run alongside on every plate

The probe-substrate designation is retained with the values because CYP3A4 inhibition may be substrate-dependent. A curve measured with this reaction is not a general potency value for every CYP3A4 substrate.

4 Matrix and species

Table 4. Matrix and readout definitions

Assessment	Biological matrix	Species	Readout
Five-isoform DDI screen	Liver microsomes	Human	% inhibition at 10 μ M
CYP3A4 inhibition curves	Liver microsomal probe-reaction context	Human	IC50 in μ M; midazolam 1'-hydroxylation

No rat inhibition table or whole-cell induction result is supplied. These are enzyme-inhibition observations in a human in vitro context. They do not establish the concentration of test article at the enzyme in a patient.

5 Accession-lot and parent identity

The controlling accession is M16/BIO/02193. Complete TP04 identifiers connect the inhibition screen, the confirmation table and the companion assay worksheet. The submitted parent is the analyte designation used throughout.

The comparison assumes consistent test article identity between screen and curve work. A later chemistry change must retain a predecessor link while receiving its own compound and accession identity. The present results cannot be transferred to a new cap or branch without measurement.

6 Screen procedure and interpretation

The panel measures reduction of probe-substrate activity relative to the assay reference at 10 μ M test article. Because the test concentration is fixed, the screen is useful for prioritization but does not supply a full concentration-response relationship.

The five isoforms are retained separately. A concern in one column is not proof of equivalent inhibition in the others. The CYP3A4 confirmation is a targeted follow-up and does not replace the remaining screen columns.

7 Concentration-response procedure

The CYP3A4 follow-up assesses activity across concentrations and reports the resulting IC50 or limit statement. The probe is midazolam, and the reaction is 1'-hydroxylation. No alternative probe curve is supplied.

An IC50 describes inhibition within the stated assay context. No K_i , time-dependent inactivation constant or mechanistic inhibition class is derived here. Those quantities require experiments beyond this screen-and-confirmation design.

8 Reference-response benchmarks

The released reference-response values are shown with their historical comparators. The comparison supports a reading of assay behavior without introducing a numerical acceptance band absent from the supplied record.

Table 5. Reference response and historical comparison

Isoform	Current reference response (μM)	Historical comparator (μM)
CYP1A2	2.1	1.8
CYP2C9	0.48	0.35
CYP2C19	1.1	1.2
CYP2D6	0.065	0.060
CYP3A4	0.035	0.030

Current and historical values are not test article results. Their proximity or difference cannot, by itself, establish that every underlying analytical sample passed a numerical criterion.

9 Screen summary by isoform

At the 50% screen flag, the numbers of compounds above the flag are 2 for CYP1A2, 4 for CYP2C9, 5 for CYP2C19, 2 for CYP2D6 and 5 for CYP3A4.

The number above the flag in any isoform is 5. Counts across columns overlap and must not be summed as a count of distinct compounds. The benchmark identifies CYP2C19 and CYP3A4 as the most flagged isoform; that designation is not a declaration that other isoforms are free of concern.

10 CYP3A4 confirmation summary

The concentration-response release includes 18 compounds. The number with IC₅₀ below 1 μ M is 1; the number below 10 μ M is 5. These are nested categories and must not be added.

The screen percentage and curve IC₅₀ have different units and meanings. Their agreement is assessed by direction and relative ordering, not by treating a percentage as a concentration or forcing a numeric conversion between them.

11 Analytical measurement responsibilities

Javier Quintana is the mass-spectrometry lead. Analyte cards and the assay worksheet retain matrix-associated analytical details and parent assignment. The report provides released inhibition results rather than reproducing individual analytical acquisitions.

Interference with probe-product measurement can resemble inhibition if analytical response is not kept separate from enzyme activity. The final narrative therefore retains the matrix context and does not assign mechanism solely from a reduction in measured activity.

12 Analytical limits

Numerical quantitation limits, calibration ranges and raw response values are not supplied with the final tables. The report does not create them. The result release supports comparison of inhibition, not reconstruction of every probe-product measurement.

A greater-than IC50 entry indicates that the reported boundary was not crossed within the released range. It is neither zero inhibition nor an exact IC50 at the printed limit. Negative screen values likewise remain part of the signed data presentation.

13 Matrix acceptance and exceptions

Helen Marchwood is responsible for bioanalytical quality. The companion assay worksheet contains matrix acceptance checks. Reference-response benchmarks are retained above as the context supplied with this final result record.

No named sample-level exception, repeat history or numerical acceptance interval is supplied. None is inferred. The report does not equate a favorable test article profile with acceptance of all analytical samples, nor add an exclusion list that is absent from the record.

14 Comparison of screen and follow-up

For the TP04 branch decision, the confirmation work supported the screen's rank order. The value of follow-up was to test whether a single-concentration impression persisted across concentration response, rather than promote every screen flag directly to a final decision.

Confirmation remains probe-specific and CYP3A4-specific. It does not supply concentration-response results for the other isoforms. The complete panel should remain available when the client considers later chemistry choices.

15 Chemistry decision basis

Tervemere retained the low-interaction branch after the concentration-response work confirmed the screen's rank order. The branch decision is a comparative choice within this test article set; it is not a claim that all retained chemistry is without interaction potential.

No per-compound branch label is supplied beyond the table identifiers. Such labels are not invented here. Any later synthesis plan should link its branch designation to the original compound record.

16 Exposure context and DDI boundaries

The study contains no circulating exposure measurement, unbound clinical concentration or administered-dose record. An in vitro IC₅₀ cannot be converted into clinical interaction risk without that additional context.

No induction, time-dependent inhibition or transporter assessment is reported. The term DDI screen is the house name for this inhibition panel; it must not be expanded into a claim that all mechanisms of interaction have been evaluated.

17 Handoff to later work

The retained branch should preserve original identifiers in the chemistry handoff. A subsequent test article set can be compared with this screen and confirmation basis, while keeping changes in material identity explicit.

Later work should keep the probe reaction and test concentration visible when comparing results. A different assay context can answer a useful question but should not be treated as identical evidence without qualification.

18 Final conclusion

Tervemere retained the low-interaction branch after the concentration-response work confirmed the screen's rank order. Megan Larkin records this as the final study outcome for Tervemere Pharma AG under MV-2016-095.

The decision is supported within the scope of the released human microsomal inhibition data. Matrix definitions, probe specificity, limit handling and analytical qualifications remain part of any extract used for the exposure gate.

19 Report control and assumptions

This final R1 report is dated 06/16/2016. The companion MV-2016-095_AssayWorksheet_061616_v03.xlsx holds accession-lot mapping and matrix acceptance. The 06/07/2016 interim communication is superseded for closeout by this final interpretation.

- Preserve complete compound and accession identifiers.
- Do not sum overlapping isoform flag counts.
- Keep nested curve counts distinct.
- Retain bound signs and negative screen values.
- Keep the midazolam probe reaction with CYP3A4 IC50 results.
- Do not extend this inhibition release to clinical risk or untested chemistry.

Name	Role	Signature	Date
Megan Larkin	DMPK Study Director	<u>M Larkin</u>	06/16/2016
Javier Quintana	Mass-Spectrometry Manager	<u>J Quintana</u>	06/16/2016
Helen Marchwood	Bioanalytical Quality Manager	<u>H Marchwood</u>	06/16/2016